Case 3 - R&D; / Trade Secret Invention Strategy

CONFIDENTIAL - SYNTHETIC BOARD / TECHNICAL REVIEW MATERIAL

Synthetic board-level package: confidential R&D;, manufacturing know-how, controls, existing patent coverage, disclosure risk, and invention strategy.

Board strategy context

HelioForge Systems develops high-density liquid-cooled AI compute modules. Management believes the company's most valuable intellectual property may not be captured by its filed patents because several performance improvements emerged as manufacturing and control-system know-how. The board wants an integrated R&D; / trade-secret / invention strategy before the next financing.

1. Existing patent portfolio - simplified claims

Family	Core claimed concept
HF-101	Cold plate with parallel microchannels beneath accelerator package.
HF-118	Rack manifold with redundant pumps and pressure sensors.
HF-133	Controller that adjusts coolant flow as a function of measured package temperature.

2. Confidential R&D; notebook excerpts

Experiment 26-041: During rapid workload transitions, package hotspot temperature overshoots even when average coolant flow is increased. Engineer L. Chen observed that the derivative of GPU power demand predicts the hotspot 180-260 ms before the package temperature sensor responds.

Experiment 26-052: A controller using workload power derivative plus inlet-fluid compressibility estimate reduced 99.9th-percentile hotspot overshoot by 37% relative to temperature-feedback control. The effect disappeared when the accumulator pre-charge was changed by more than 12%.

Experiment 26-061: Manufacturing found that intentionally varying microchannel width across the cold plate by +/-6% counteracted package-level heat-flux asymmetry. The CAD variation is generated from each package's measured post-assembly thermal map, not from a fixed geometry.

3. Manufacturing process notes

The adaptive channel-width map is generated after package attach. A proprietary script converts thermal-map coordinates into laser-machining offsets. Operators see only a job identifier; the transformation coefficients are stored in an access-controlled build server. Yield improved from 91.2% to 97.8% after introduction.

A supplier receives final machining coordinates but not the thermal map, algorithm, or coefficients. The supplier contract prohibits reverse engineering and reuse of job files. Coordinates are unique to each unit and retained by the supplier for 30 days.

4. Controls / firmware notes

Feature	Current implementation	Publicly documented?
Temperature feedback	PID using package temperature	Yes - patent HF-133
Power-derivative feed-forward	Predictive term from accelerator telemetry	No
Compressibility estimator	Online estimate from pressure-wave response	No
Accumulator pre-charge coupling	Calibration table links estimator to pre-charge	No
Per-unit channel geometry	Thermal-map-derived machining transformation	No

5. Commercial and reverse-engineering observations

Customers can measure external coolant pressure, inlet/outlet temperature, and application performance. They cannot directly inspect firmware because secure boot and signed updates are enabled. Destructive teardown reveals the physical channel geometry, although inferring the per-unit generation algorithm would require comparing multiple units with their pre-machining thermal maps, which customers do not possess.

A competitor recently advertised 'predictive liquid cooling using workload telemetry' but has not disclosed its algorithm. No known competitor advertises per-package adaptive cold-plate geometry.

6. Internal invention disclosure status

No invention disclosure has been filed for the feed-forward controller, compressibility estimator, accumulator interaction, or per-unit geometry-generation method. Engineering has treated all four as implementation details. Sales presentations disclose that HelioForge achieves 'predictive cooling' and 'factory-calibrated cold plates' but do not disclose the mechanisms.

Challenge prompt

> **Act as a board-level R&D; / Trade Secret Invention Strategy team. Using only this confidential package, identify technical concepts that appear materially distinct from the existing patent claims; separate potentially patentable inventions from know-how that may be more valuable as trade secrets; identify combinations that may be more defensible than individual features; assess what is exposed to reverse engineering or accidental disclosure; and propose the next diligence experiments and documentation steps. Cite the evidence supporting every conclusion and explicitly flag where outside prior-art or legal analysis is required.**